

# On Fairness, Explainability, and Transparency in Computer Vision, Robotics, and Aviation

Jaagat Prashar

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## 1 Introduction

Since the discussion of AI at the 1956 Dartmouth Summer Research Project, the fields of artificial intelligence (AI) and automation have revolutionized society. From self-driving cars to Open AI's ChatGPT (Generative Pre-Trained Transformer), AI has seamlessly seeped into our everyday lives, accelerating the rate at which we accomplish tasks and fundamentally improving the accessibility of technology to humans (MIT Technology Review).

Broadly, AI's applications in fields such as healthcare and education have transformed the quality of life of many. Take, for instance, the invention of AlphaFold by Google DeepMind, an AI system that enables scientific researchers to accurately predict the 3D structure of a given protein based upon the respective amino acid sequence (AlphaFold). Additionally, through advancements such as AI tutors (e.g., Khan Academy's Khanmigo) and educational Chatbots, education has become more accessible than ever, providing students with an avenue to ask questions, especially when limited teaching resources may be available (Khanmigo).

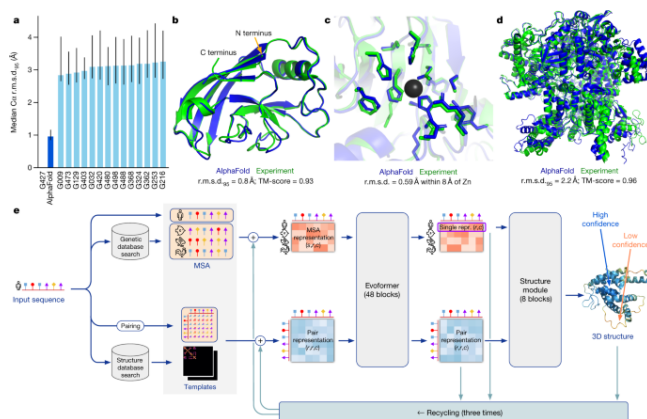


Figure 1: AI Pipeline for AlphaFold by Google DeepMind; AlphaFold leverages multiple-sequence alignment to predict protein structures.

Most notably, the influence of AI and automation in computer vision, robotics, and aviation has played significant roles in promoting the safety of humanity, enabling convenience in our daily lives, and expediting the rate at which we are able to complete tasks. For example, Waymo LLC, an autonomous driving company, has made significant advancements in the domain of autonomous driving with the intent of making transportation more accessible, efficient, and safe. According to Waymo's safety reports, Waymo has enabled 81% fewer airbag deployment crashes, 78% fewer injury-related crashes, and 62% fewer police-reported crashes, significantly enhancing safety on the road (Waymo Safety). Additionally, a recent startup named Figure AI has deployed their humanoid robot BotX in BMW factories to minimize the risk that factory workers are often subjected to in production lines (Figure AI). Lastly, in the field of aviation, developments such as ACAS X and the Aviation Safety Reporting System have profoundly improved the safety of millions of people who travel on a daily basis (SKYbrary).

Nonetheless, although AI and automation have made significant advancements in humanity, concerns regarding the fairness, explainability, and transparency of such decision-making algorithms have shed light on the ethics of deploying AI systems. For example, in American mathematician and data scientist Cathy O'Neil's TED talk *The era of blind faith in big data must end*, O'Neil passionately defines algorithms as "weapons of math destruction" since they are effectively "opinions embedded in code" (TED). Specifically, she references the example of New York employing a cryptic value-added model that led to the unfortunate firing of school

teachers despite positive reviews from students and parents. More strikingly, she references the example of a blind orchestra audition, a canonical example pertinent to algorithmic bias, revealing that when blind auditions were conducted for an orchestra, approximately 5 times more women were considered. The sheer statistic of 5 times more women being selected for the orchestra is shocking, depicting just how critical these algorithms can be in propagating gender biases and influencing people's lives.

In this report, I will explore the realms of fairness, explainability, and transparency with an emphasis on computer vision, robotics, and aviation to highlight existing limitations in such fields. I will elaborate on how we can deploy algorithms that are trustworthy, safe, and accurate to strategically benefit humanity.

## 2 Fairness, Explainability, and Transparency: Overview and Case Studies

First and foremost, before exploring the applications of fairness, explainability, and transparency in computer vision, robotics, and aviation, it is vital to define fairness, explainability, and transparency:

1. Algorithmic fairness refers to whether or not the algorithm in question is establishing biased outcomes. Fairness seeks to ensure that algorithms establish decisions that are not biased against a given demographic, gender, or sexual orientation, for example.

2. Explainability references the ability to reason behind a model's decision. With the prevalence of AI "Black Boxes," it has become increasingly important to interpret the internal decisions that lead to an algorithm's final output.

3. Transparency corresponds to how accessible the algorithmic model and data used for training/testing are. The goal of transparency is to increase trust by ensuring that people are well-informed about how AI models are designed, trained, and implemented.

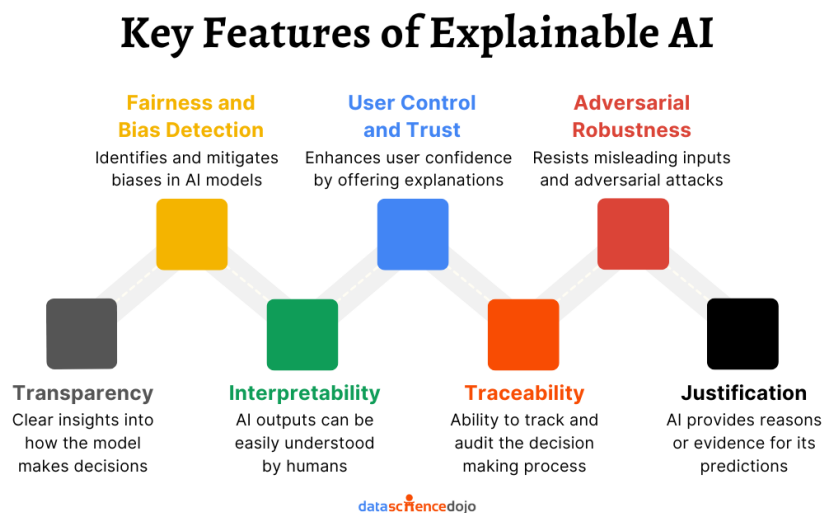


Figure 2: The fundamentals involved in ensuring AI is fair, transparent, and explainable.

With AI increasingly governing our day-to-day lives through applications such as Chatbots and autonomous driving, the public has raised concerns for algorithms to be as fair as possible. Inherently, for AI models to establish predictions in the first place, they must be fairly trained on large, representative datasets – a keystone point that serves as a foundation for case studies we will explore:

### 2.1 Computer Vision - ImageNet Roulette

In an attempt to illustrate the importance of algorithmic bias, AI researcher Kate Crawford and artist Trevor Paglen created the ImageNet Roulette, a project that significantly exposed striking biases in the ImageNet dataset, a massive image database that is often deployed for computer vision model tasks (e.g., object detection, image classification, etc.). The goal of the project was to enable users to upload their photo to the interface and as a result, get a label classifying the inputted face image. Upon the release of ImageNet Roulette, social media posts concerning the ethics of ImageNet had proliferated across platforms such as X, highlighting how significantly biased ImageNet was. Specifically, in the article *The viral selfie app ImageNet Roulette seemed fun – until it called me a racist slur*, journalist Julia Carrie Wong revealed how the interface had labeled her with a racist slur (The Guardian).

Although ImageNet was developed with the intention to support the decision-making capabilities of computer vision algorithms since AI models fundamentally rely on data to learn representations and patterns, we must consider the data we are using to train such models and how biases existing in such data can further propagate discriminatory, biased decisions. Following the announcement of ImageNet Roulette, ImageNet had removed 438 people categories and 600,040 associated images that they deemed unsafe (Artforum). More importantly, the researchers behind ImageNet declared intent to enable people to communicate direct reports in the event that they come across offensive labels or terminology.

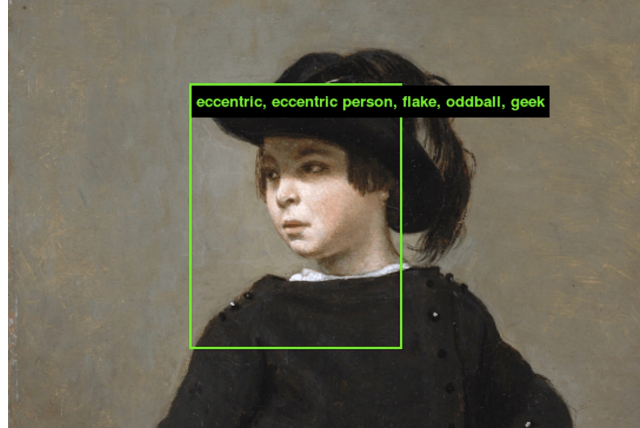


Figure 3: Example output from ImageNet Roulette.

In essence, ImageNet Roulette highlights the core principles of fairness, explainability, and transparency by emphasizing how data serves as a driving force for algorithmic decisions; in order for algorithmic decisions to be fair, we must first ensure that the data we are using to train such models are indeed fair and representative as well. Although ImageNet Roulette brought a greater focus on how AI scientists must consider the fairness of their data in order to promote transparency and explainability in the algorithmic decision-making process, debates surrounding the inevitability of bias in data and humans have continued to raise concerns. Thus, an existing limitation is the fact that large amounts of data are needed to improve the accuracy of AI models, yet it is often difficult to obtain large datasets that are fair. Additionally, humans are fundamentally biased and therefore have the capacity to propagate biases when developing AI models (e.g., labeling). We will discuss both of these limitations later in the report.

## 2.2 Computer Vision - Waymo

Waymo is an autonomous driving company that employs computer vision algorithms to enable sustainable, safe autonomous driving. Holistically, Waymo has demonstrated better vehicle safety performance compared to vehicles driven by humans, significantly reducing injuries and property damage. With road accidents constituting one of the top leading causes of death, it is clear that Waymo's autonomous driving algorithm holds an immense capacity to promote the safety of human lives. Nonetheless, Waymo – as is everything – is not perfect. Between the years of 2021 and 2024, there were reportedly 696 Waymo accidents (not necessarily caused by a Waymo vehicle), highlighting potential concerns regarding how safe Waymo's technology truly is. For instance, in February 2024, two Waymo cars had both hit a pickup truck due to a misrepresentation in Waymo's computer vision's algorithm. Consequently, Waymo, after holding critical discussions regarding the safety of Waymo with the National Highway Traffic Safety Administration (NHTSA), had issued a recall report, demonstrating the need for its software to be improved. To this end, Waymo plays a pivotal role in highlighting the need for fairness, transparency, and explainability in software development.

The decision to issue a recall report to investigate why their computer vision algorithm incorrectly detected the visual scenario of the pickup truck and the subsequent action to fix their software demonstrates Waymo's commitment to transparency. In an effort to promote fairness, Waymo has continued to make active efforts to deploy its product in diverse locations such as San Jose, CA and Austin, TX. By doing so, Waymo aims to develop a diverse dataset that is capable of enabling representative decisions. Lastly, Waymo promotes the notion of explainable AI by ensuring that its user can physically see the outputs from the model (i.e., a person walking, skateboarding, biking, etc.).

Although Waymo isn't perfect, it significantly promotes the safety of human lives and has in fact, outperformed the safety benchmark of human drivers, who are often more susceptible to making mistakes due to being tired or irresponsible (Waymo).

## 2.3 Robotics - Boston Dynamics

Boston Dynamics is a robotics design company that is rooted in the mission of deploying mobile robots that are capable of enhancing and automating human tasks. Through the development of surprisingly mobile and efficient robotic systems that can accurately mimic human dexterity and movements, concerns have been raised about the potential weaponization of such robots. In fact, an increasing number of news headlines have deemed the robot to be “terrifying” (Boston.com News). Consequently, Boston Dynamics has promoted transparency by clearly establishing an ethical guidelines framework that outlines the company’s mission and how they intend to use the robot. Namely, the company has declared that they are “motivated by curiosity and respect for humans and animals,” “prioritize the human element in human-robot partnerships,” “build trustworthy robots,” and – most importantly – that “we will not weaponize our robots,” to name a few (Boston Dynamics). Despite establishing such clear guidelines depicting the company’s harmless intent to promote the usage of robotic systems for societal good, the public has opposed such innovation from the company; the New York City Council very recently established a bill to prohibit the NYPD from using “armed robots” – albeit Boston Dynamics has clearly stated they wouldn’t permit such activity (ABC).

However, in cities such as LA, the police force – LAPD – has employed Boston Dynamics’s robot dog *Spot*. Consequently, in order to further promote transparency in Spot’s usage, the LAPD is obligated to provide quarterly reports with specific details about exactly how Spot is being utilized. Furthermore, in an effort to promote fairness and explainability in Spot’s algorithm, Boston Dynamics has published whitepapers (detailed industry reports) that highlight how the company is addressing improvements (i.e., acoustic vibration sensing, security, case unloading, etc). Moreover, a significant effort has been made to open-source Spot’s SDK (Software Development Kit), thereby making Spot’s algorithm accessible to researchers across the globe.

## 2.4 Aviation

In the domain of aviation, ACAS X has revolutionized the safety of air travel for millions of people. Foundationally, ACAS X relies on principles such as dynamic programming and probabilistic models to interpret the most statistically likely representation of an aircraft’s position. Holistically, ACAS X has improved the safety of aviation by reducing unnecessary advisories, enabling future surveillance, and preventing mid-air collisions. The reliability and accuracy of ACAS X is evident through its role in transparency, reliability, and fairness. Whereas deep neural network algorithms tend to often be “black boxes,” ACAS X’s role in enabling collision avoidance acknowledges the notion of decision making under dynamic uncertainty through sensor measurements, logic lookup tables, probabilistic distributions, and Markov decision processes, to name a few. The detailed implementation of ACAS X is publicly available through reports that delineate the limitations in TCAS and how ACAS X enables a greater reduction in collision risk (specifically by 47%), thereby increasing the transparency and explainability of the system. Moreover, the explainability of the system is delineated through the process by which it performs state estimation and action selection every second.

Furthermore, through the establishment of systems such as the Aviation Safety Reporting System (ASRS), which was established in 1976 by NASA and the FAA with the mission of reducing risk accidents, there has been a significant rise in the transparency of safety and data integrity behind aircraft incidents. The goal of the ASRS is to review confidential reports, analyze relevant safety data, and then provide a critical assessment on how future incidents can be prevented. By analyzing voluntarily submitted reports regarding aircraft incidents, ASRS not only enables greater transparency as to why such incidents are occurring to begin with, but promotes greater explainability and fairness in assessing how we can prevent such incidents from occurring again in the future.

## 3 Existing Limitations and Future Work

Whether it’s robotics or computer vision algorithms enabling object classification, data plays a fundamental role in training models to establish robust, sustainable decisions. Nonetheless, it is vital to realize that humans are inevitably biased and thus propagate biased data through social media and the Internet. In order to ensure that we are training algorithmically fair, transparent, and explainable AI algorithms, we must first ensure that the medium by which we collect data is fair. Because unbiased data tends to be limited, scientists have begun to explore how we can enable synthetic data generation through generative adversarial networks (NVIDIA).

Furthermore, given that businesses tend to emphasize proprietary innovation and avoid participating in open-source, the notion of “algorithmic accountability” has become increasingly relevant, whereby constituents should be able to democratically voice if an algorithm is fair or not (TED). In the rise of AI advancements and proprietary algorithms, we must establish benchmarks for what constitutes fairness, explainability, and transparency insofar that AI algorithms that are proprietary do not pose a risk to humanity (as was seen with the firing of NYC teachers). While in the short-term we can establish mostly accurate decisions, if we don’t address the ethics of deployed algorithms, we risk propagating irreversible biases in the long-term.

Lastly, in an interview with Hugging Face co-founder Thomas Wolf, I learned that a striking limitation lies in the gap between scientific academia and policymakers. More often than not, policymakers aren't able to understand the technical jargon that is involved in training ML algorithms and ensuring that they establish fair decisions. Thus, as a society, we must establish incentives for open-sourcing algorithms and communicate them in a clear, effective manner that enables people from all disciplines to contribute to and understand existing software – after all, if one does not understand a software, how can they truly assess its fairness, explainability, or transparency?

## 4 Conclusion

Whether it's being prompted with Google Gemini's "AI Overview" everytime you enter a query in Google or unlocking your device through facial recognition, AI is inescapable – in a society where AI will no longer be a foreign term, but foundational to growing developments, we must ensure that we not only deploy robust algorithms, but ensure that they are fair, transparent, and explainable in their decisions.

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